



GS FOUNDATION 2024-25

GEOGRAPHY - 5

Atmospheric Circulation

Jet Streams

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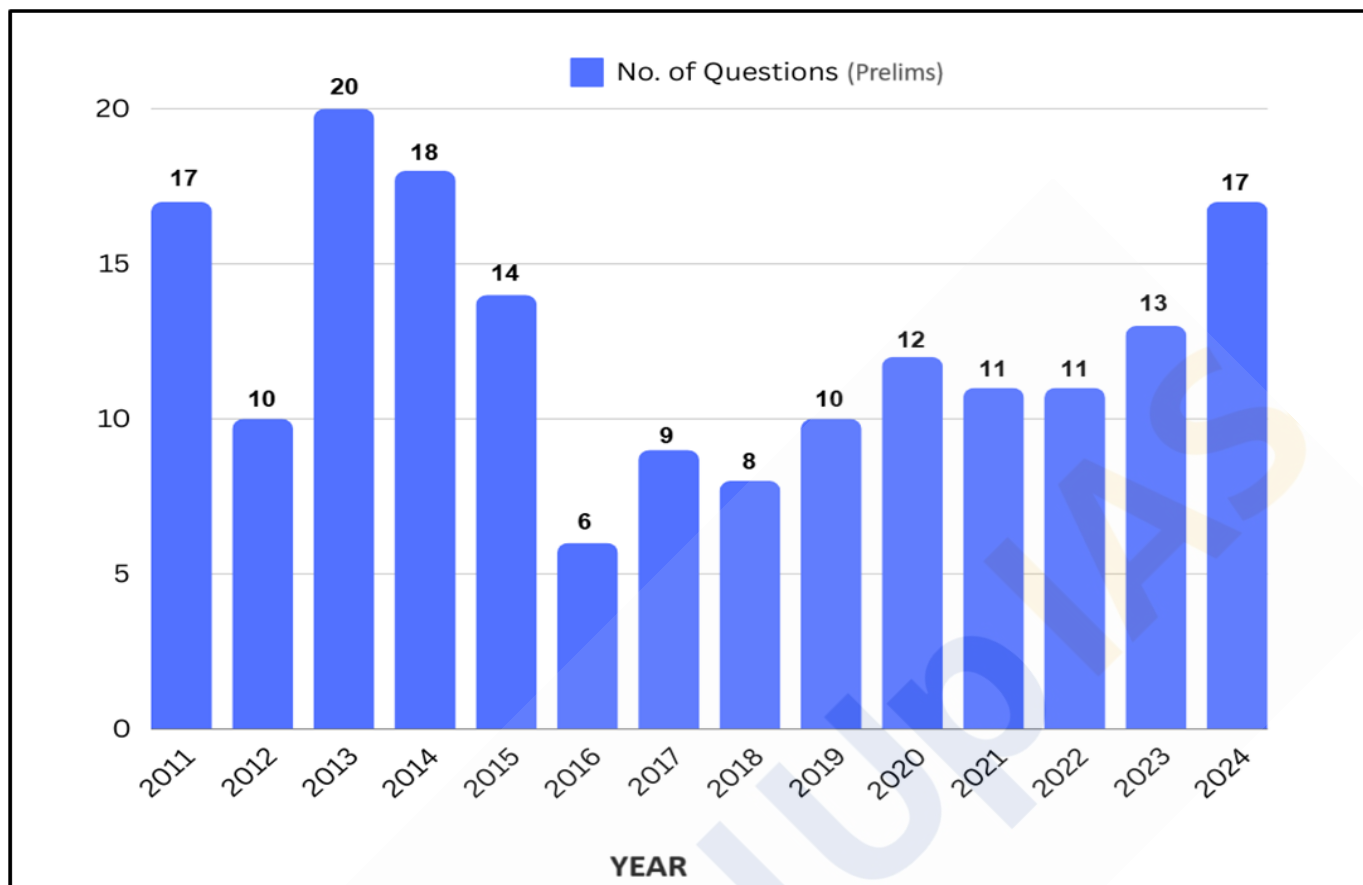
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Atmospheric Circulation

Introduction:

Atmospheric circulation refers to the **movement of air due to pressure gradients**, ranging from local to global scales and from daily to annual patterns. This movement occurs both over the Earth's surface (including land and oceans) and above it, encompassing spatial, temporal, and altitudinal components.

Types of Atmospheric Circulation:

- **Primary or General Circulation** - This includes the **planetary wind systems** related to global pressure belts, such as **trade winds, westerlies, and polar winds**. It also encompasses variants of global atmospheric circulation like the **Jet stream** and **Walker circulation**, along with the **Southern Oscillation**.
- **Secondary Circulation** - This consists of **cyclones, anticyclones**, and seasonal circulations like **monsoons and air masses**.
- **Tertiary Circulation** - This includes **local winds** such as the Harmattan, Chinook, Sirocco, Norwester, Buran, and diurnal winds like land and sea breezes.

General Circulation of the Atmosphere:

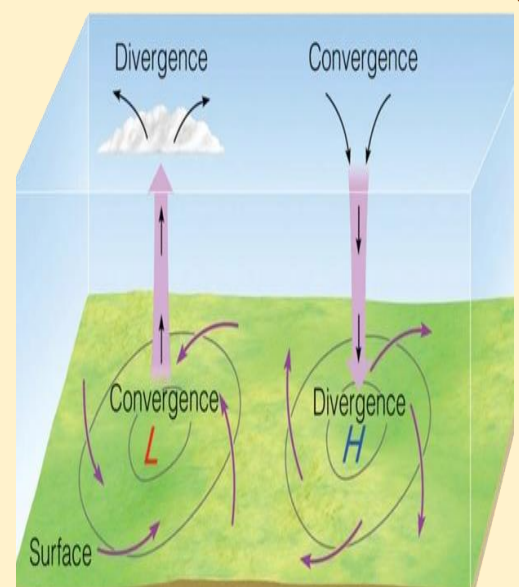
General circulation refers to the **large-scale motion of the atmosphere** involving the entire globe. Its **primary functions include:**

- **Transporting heat** from surplus areas to deficit areas, attempting to **equalize the distribution of heat energy** over Earth's surface, including both land and oceans.
- **Transporting moisture** from oceans to continents, facilitating **cloud formation and precipitation**, thereby maintaining the hydrological cycle on a global scale.

The primary cause of atmospheric motion and general circulation is the **spatial imbalance of heat energy**. The second controlling factor is the **angular momentum of Earth and its atmosphere**, which is gravitationally attached to the Earth and rotates with it. Heat and angular momentum are transported vertically and horizontally through different mechanisms.

Some Basic Concepts

- **Zone of Convergence:** It is a zone where there is a **net inflow of air**. **Low pressure conditions** create zones of convergence.
- **Zone of Divergence:** It is a zone where there is a **net outflow of air**. **High pressure conditions** create zones of divergence.
- For **every Zone of Convergence** near **Earth surface** (in lower Troposphere), there is a **Zone of Divergence** in the **upper Troposphere**.
- For **every Zone of Divergence** near **Earth surface** (in lower Troposphere), there is a **Zone of Convergence** in the **upper Troposphere**.



Models of General Circulation:

The first step of any scientific enquiry is model building. By the term '**model**' we mean a **simplified version of a complex reality**.

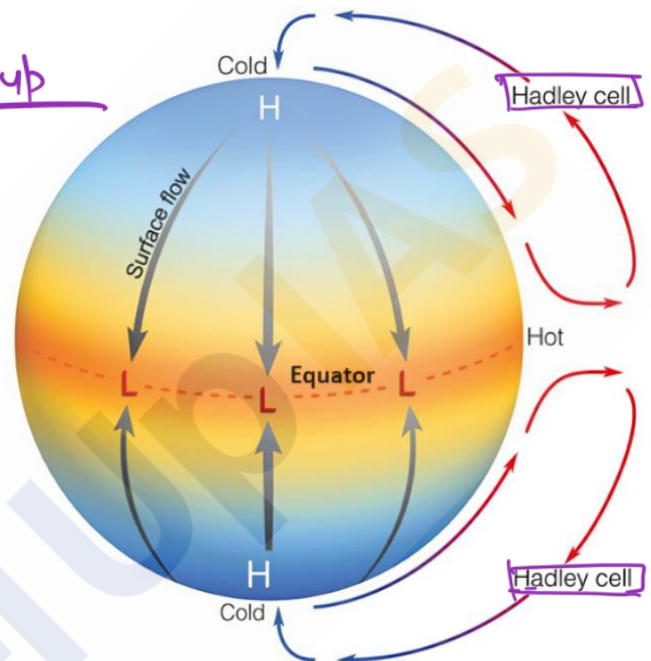
Thermal Model or Hadley's Model or Unicellular Model:

- The term "**cell**" refers to the **cycling of air**. The Unicellular model, proposed by **George Hadley in 1735**, operates on a global scale but is divided at the equator, with one cell in each hemisphere.

- Hadley's model was based on some assumptions:

- Static Earth (no rotation).
- No apparent shift of the sun's position.

- Hadley's model was based on the idea that **air in the equatorial region, heated and expanded** due to higher temperatures, **rises and moves aloft to the tropopause**. This cool air then moves towards the poles, where it **sinks** due to higher density, creating **high-pressure belts around the poles**. This model proposed that cold air sinks at the poles while warm air rises at the equator, forming the '**Hadley cell.**'

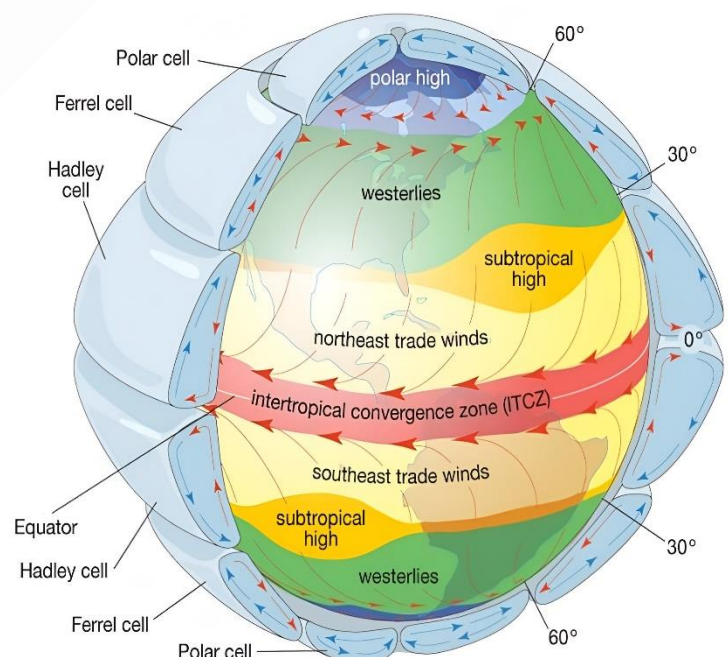


Evaluation:

- Hadley's model, though initially well-accepted, is now considered **outdated**. It assumed a static Earth and did not account for factors such as the apparent shift of the sun's position, Earth's rotation, the Coriolis effect, and the role of land and water distribution in heat transfer. Research has since shown that atmospheric circulation operates through a Three-Cell model instead.

Tri-Cellular Atmospheric Circulation Model:

- This model was given by '**William Ferrel**' in **1856**. It introduces **three interlocking cells** to explain the global wind system.
- Ferrel's model was based on some assumptions:
 - Flat and homogeneous Earth's surface (either land or water).
 - Perfectly spherical Earth.
 - Considered both rotation and Coriolis force of Earth.
 - Sun being overhead at the equator (equinox position).



Hadley Cell:

- The **Hadley cell** or **Tropical cell**, exists between the **equator and 30°** north and south latitudes. It is **thermally induced** cell.
- **At the Inter Tropical Convergence Zone (ITCZ), air rises due to convection currents driven by low pressure.** This low-pressure results from high insolation, causing **winds from the tropics to converge** at this zone. The **converged air rises** along with the convective cell, reaching the top of the troposphere up to an altitude of 14 km and then moving towards the poles.
- As the air moves towards the poles, it **accumulates at about 30° N and S**. Part of this **accumulated air sinks** to the ground, forming a **subtropical high**. Another reason for this sinking is the cooling of the air as it reaches 30° N and S latitudes.
- Near the land surface, the **air flows back towards the equator as easterlies**. These easterlies from either side of the equator converge in the ITCZ. The circulation of air from the surface upwards and vice-versa creates cells. In the tropics, this type of cell is known as a **Hadley Cell**.

Ferrel Cell:

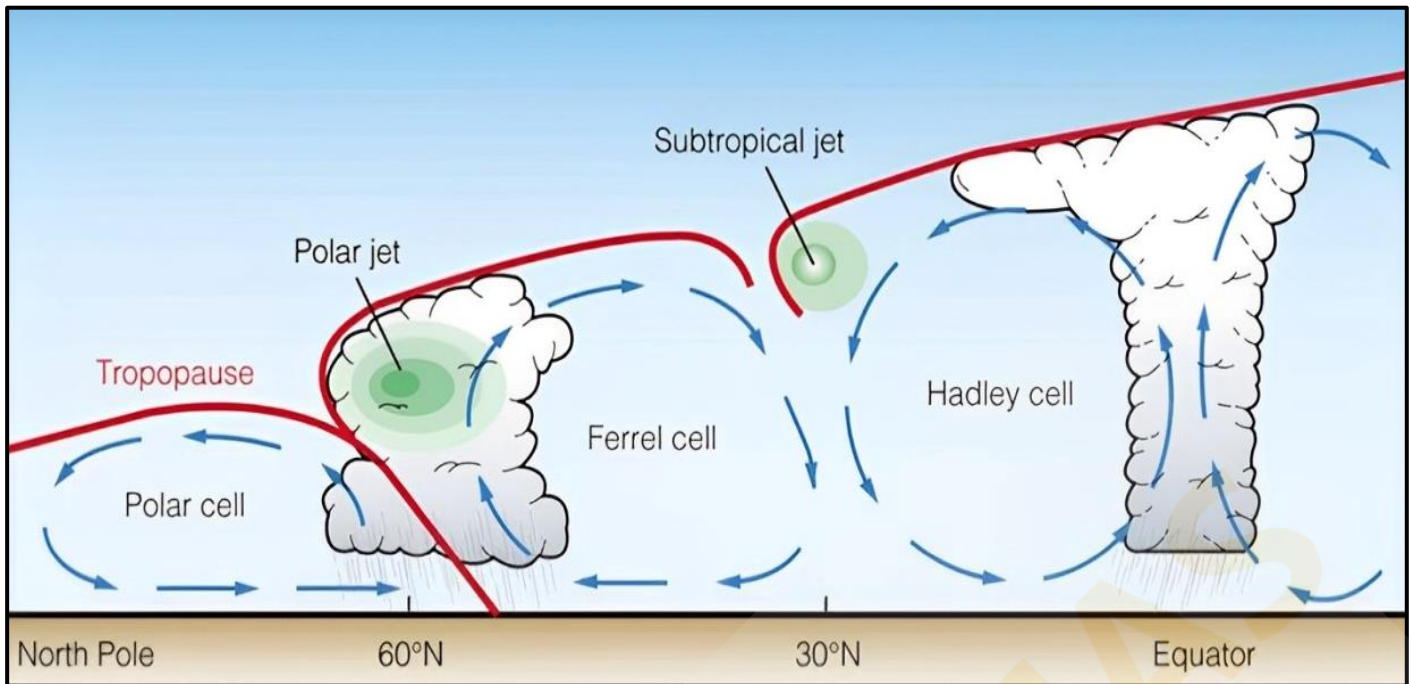
- The **Ferrel cell** or **Mid-latitude cell**, exists between **30° and 60°** north and south latitudes. It is **dynamically induced** by the interaction between the Hadley and Polar cells.
- **Winds from the subtropical high-pressure zones move towards both the equator and the poles.** Before reaching the poles, another **low-pressure zone develops around 60° north and south latitudes**. This occurs because the **poles are extremely cold, making it natural for high pressure** to form around them. Therefore, between the two high pressure zones—subtropical and polar—a low pressure area naturally arises.
- In this low-pressure belt, rising air reaches near the tropopause and is diverted towards both the poles and the subtropical regions.
- This circulation pattern, created by the movement of air from the subtropical high to the subpolar low, then upward, diverted towards the subtropical high, and finally sinking, is known as the **Ferrel cell**.

Polar Cell:

- The polar cell develops between **60° latitude and the poles in both hemispheres**. Polar areas are characterized by **thermally induced** high-pressure belts due to the excessively cold conditions.
- In these regions, the **temperature is very low, the air is dry**, and the intermolecular space within the air is minimal, resulting in high air density. Consequently, this **dense air tends to remain near the surface**.
- Cold winds, known as **polar easterlies, blow from the poles towards the sub-polar low-pressure areas**. At the sub-polar low, these winds converge with the westerlies at the polar fronts, causing the air to be lifted. This lifted air then moves towards the poles near the tropopause, completing the circulation pattern known as the **polar cell**.

Evaluation:

- The Tri-Cellular model is an **improvement over the Hadley model**. It is **easy to understand** and explains global planetary winds and broad latitudinal climate patterns. However, it has **limitations, such as not accounting for local winds during monsoons** and assuming vertical cells operate up to the upper troposphere without discussing the nature of upper tropospheric circulation.
- Since, the **Earth surface is not uniform**, therefore we do not have uniform pressure belts in nature. Instead of pressure belts we have systems of low and high **Pressure cells** (large area with more or less uniform low or high pressure).



Jet Streams

Introduction:

Jet streams are **fast-flowing, narrow bands of strong wind** that generally blow from **west to east in both hemispheres**. They are generally located in **upper layers of the troposphere**. They play a crucial role in influencing weather patterns and climate by directing the movement of air masses and storm systems across the globe.

Characteristics of Jet Streams:

- **Circumpolar:** Jet streams are winds that **circle around the earth** with poles as their centers.
- **Location and Altitude:** Jet streams are usually found between **9 to 16 kilometers** (30,000 to 52,000 feet) above the Earth's surface, near the tropopause. Their position varies with the seasons, shifting closer to the equator in the winter and towards the poles in the summer.
- **Speed:** The speed of jet streams can range from **120 to 500 kilometers per hour** (70 to 310 miles per hour). They are **stronger in the winter** when the temperature contrast between the polar and tropical air masses is greater. The **greatest wind speeds occur in the center** of the jet stream, with velocities decreasing away from it.
- **Width and Depth:** Jet streams are generally **thousands of kilometers long, a few hundred kilometers wide**, and a few kilometers deep. Despite their narrowness, they have a significant impact on the weather.
- **Direction:** Jet streams flow from **west to east due to the Earth's rotation** (Coriolis effect) and the temperature contrast between different air.
- **Meandering:** Encircle the globe following a **curved path with three-dimensional flow**, developing crests and troughs.



Formation of Jet Streams:

Jet streams form primarily due to the **differences in temperature and pressure** between contrasting air masses. The Earth's rotation also influences their formation. Key factors include:

- **Temperature Gradient:** The stark contrast between cold polar air and warm tropical air creates a steep temperature gradient, leading to the development of jet streams.
- **Pressure Gradient:** Differences in **pressure at various latitudes** contribute to the formation and strength of jet streams.
- **Coriolis Effect:** The rotation of the Earth causes moving air to deflect to the right in the Northern Hemisphere and to the left in the Southern Hemisphere, shaping the flow of jet streams.

Types of Jet Streams:

Permanent Jet Streams:

- Subtropical Westerly Jet Streams
- Polar Front Jet Streams

Temporary Jet Streams:

- Tropical Easterly Jet Streams or African Easterly Jet Streams
- Somali Jets

Other Types of Jet Streams:

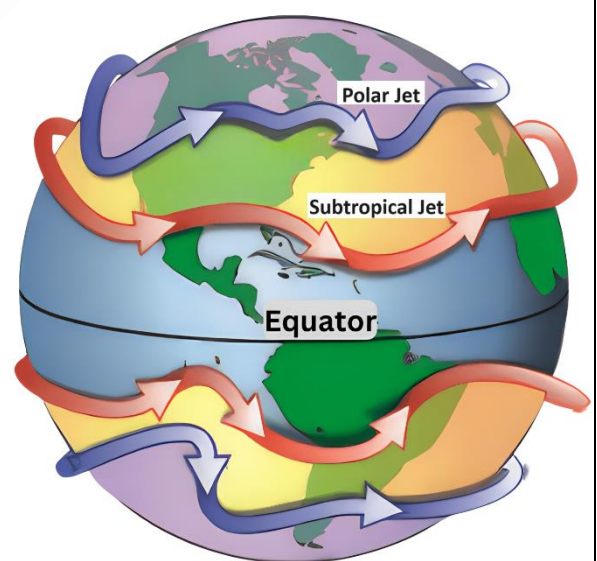
- Polar Night Jet Streams
- Barrier Jet Streams
- Coastal Low-Level Jet Streams
- Valley Exit Jet Streams

Subtropical Westerly Jet Streams:

- The sub-tropical jet stream is produced by the **earth's rotation** (Coriolis force) and **temperature contrast between tropical and sub - tropical regions**.

- At the equator, the rotation results in the greatest atmospheric velocity. Consequently, the rising air, which spreads out northward and southward, moves faster than the latitudes it flows over. This air is deflected to the right in the Northern Hemisphere and to the left in the Southern Hemisphere, becoming concentrated around **30° latitude as the Subtropical Jet Streams**.

- During **winter**, the **Subtropical Jet Streams** is **nearly continuous in both hemispheres**. In the **southern hemisphere**, it exists **year-round**, while in the **northern hemisphere**, it becomes **intermittent during summer**, migrating northwards.
- The Subtropical Jet Streams can be **temporarily displaced** when **strong mid-latitude troughs** extend into **subtropical latitudes**. These displacements can cause the subtropical jet to **merge with the Polar Front Jet**. This interaction is related to phenomena like **cloudbursts**, which will be studied in the context of Indian Monsoons.



Polar Front Jet Streams:

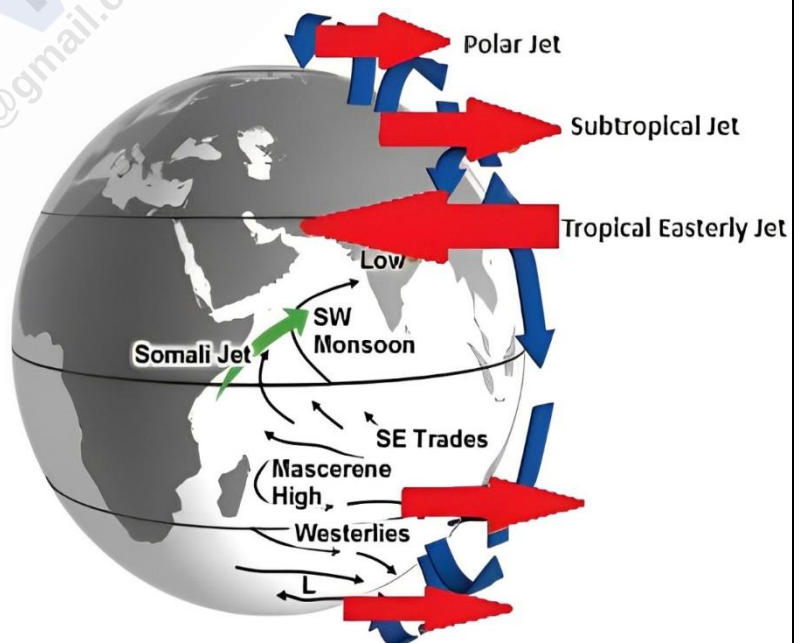
- Polar front jet streams are **formed above the convergence zone of the surface polar cold air mass and the tropical warm air mass**, typically between **40 degrees and 60 degrees latitude**. This convergence results in a steep thermal gradient due to the collision of two contrasting air masses.
- The **position** of the polar front jet stream is **more variable than that of the subtropical jet stream**. During the **summer, the jet stream shifts towards the poles**, while in the **winter, it moves towards the equator**. In winter, the polar front jet stream becomes particularly strong and continuous, significantly influencing the climates of regions near 60 degrees latitude.
- The polar front jet stream plays a **crucial role in determining the path, speed, and intensity of temperate cyclones**. Its movement and strength can greatly affect **weather patterns and climatic conditions** in the **mid-latitudes**, making it a key factor in meteorological studies and forecasts.

Tropical Easterly Jet Streams or African Easterly Jet Streams:

- The Tropical Easterly Jet is a prominent upper-level wind phenomenon occurring **near the tropics**, specifically during the **summer months** in the **Northern Hemisphere**.
- **Originating in late June and extending until early October**, this upper-atmosphere **easterly flow** spans from **Southeast Asia to Africa**, found near **between 5° and 25°N**. It reaches its peak strength approximately 15 kilometers above the Earth's surface over the Indian Ocean, boasting wind speeds that can reach up to 40 meters per second.
- The formation of this jet stream is closely tied to the **heating of the Tibetan Plateau** and the shifting dynamics of the Subtropical Jet as it moves northward from the Himalayas. Importantly, these jets play a **crucial role in influencing the Indian monsoon patterns**. It induces secondary circulations that enhance convection over south India and nearby ocean.

Somali Jet Streams:

- The Somali Jet, a prominent tropical **low-level jet stream**, is observed primarily during the **summer months over northern Madagascar and off the coast of Somalia**.
- During its **peak intensity from June to August**, this jet stream **splits into two branches** - northern branch meets the west coast of India near 17°N, and southern branch that moves eastward just south of the Indian peninsula.
- Throughout the summer season, from **June to September**, the Somali Jet remains relatively **stable** before **gradually shifting southward** towards the southern Indian Ocean during the winter months.
- This regional jet stream is **crucial for the summer monsoon dynamics**, playing a vital role in transporting moisture and **contributing to the onset of monsoonal rains** across the Horn of Africa and the Arabian Sea region.



Polar Night Jet Streams:

- During the **winter months**, when **polar nights** extend over their respective hemispheres around the **60° latitude**, the polar-night jet stream forms. This **high-altitude jet stream**, at **approximately 24 km**, moves more prominently compared to its summer position.
- During these dark periods, the **air above the poles becomes significantly colder than equatorial regions**, creating substantial stratospheric **pressure differences**. Combined with the Coriolis effect, these conditions generate the **polar-night jets**, which swiftly **travel eastward at an altitude of about 48 km**.
- Encircling the **polar vortex**, the **jets prevent warmer air from entering**, keeping the cold polar air isolated and increasingly frigid throughout the polar night.

Barrier Jet Streams:

- In the **low levels**, a barrier jet forms just **upstream of mountain chains**, causing the jet to be oriented **parallel to the mountains**.

Coastal Low-Level Jet Streams:

- Coastal low-level jets are associated with a **sharp contrast between high temperatures on land and lower temperatures at sea**, and they play an important role in coastal weather, causing strong coast parallel winds. These are linked to **oceanic high-pressure systems and land-based thermal lows**.

Valley Exit Jet Streams:

- A valley exit jet is a **strong, down-valley, elevated wind** that flows above the **valley's and adjacent plain's intersection**. They have the ability to travel at high speeds and heights.

Influence of Jet Streams on Weather and Climate:

Jet streams have a profound impact on weather and climate patterns by steering weather systems and influencing the distribution of temperature and precipitation. **Their effects include:**

- **Storm Tracks** - Jet streams **guide the path of storm systems**, determining their **intensity and movement**. For instance, a strong polar jet stream can drive severe winter storms across mid-latitudes.
- **Temperature Distribution** - The position of jet streams affects the **distribution of warm and cold air masses**. When a **jet stream dips southward**, it can **bring cold Arctic air into lower latitudes**, while a northward bulge can transport warm tropical air poleward.
- **Precipitation Patterns** - Jet streams influence the **development and movement of weather fronts**, which in turn affect precipitation patterns. Regions under the influence of jet streams can experience significant rainfall or drought depending on the jet stream's position. It also plays a significant role in the onset and withdrawal of monsoon winds.
- **Climate Variability** - Long-term changes in the behavior of jet streams can contribute to climate variability and change. For example, shifts in the jet stream's position and strength can affect the occurrence of extreme weather events such as heatwaves and prolonged cold spells.

Jet Streams and Aviation:

Jet streams play a critical role in aviation by affecting flight routes and times. Understanding their patterns allows airlines to optimize flight paths, reduce fuel consumption, and ensure passenger safety. Key considerations include:

- **Tailwinds and Headwinds:** Pilots use the strong tailwinds of jet streams to **reduce flight times and save fuel**. Conversely, **headwinds can increase travel time and fuel consumption**.
- **Turbulence:** Jet streams are associated with **clear-air turbulence**, which can pose challenges for aircraft. Pilots and meteorologists closely monitor jet streams to avoid turbulent areas and ensure smooth flights.

Jet Streams and Climate Change:

Climate change is influencing the behavior of jet streams, with potential implications for weather patterns and extreme events. Key impacts include:

- **Arctic Amplification** - The **rapid warming of the Arctic region**, known as Arctic amplification, is altering the temperature gradient between the poles and the equator. This can **weaken the polar jet stream**, leading to more meandering patterns and **increased frequency of extreme weather events**.
- **Jet Stream Shifts** - Changes in the position and strength of jet streams can affect regional climates. For example, a **northward shift of the polar jet stream can result in warmer and drier conditions** in some regions, while a **southward shift can bring cooler and wetter conditions**.
- **Extreme Weather Events** - Altered jet stream patterns are linked to the increased frequency and intensity of extreme weather events such as **heatwaves, cold spells, heavy rainfall, and droughts**.

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